Mohammad Elham Nejrabi

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EDUCATION

University of California, Davis — BS Electrical Engineering — Expected June 2028
Diablo Valley College — AS Computer Science, GPA 3.7 — Graduated May 2026

ENGINEERING EXPERIENCE

Assistant Manager, Domino's Pizza — *Nov 2021 – Present*
Operations Management — Walnut Creek, CA

- Managed daily operations coordinating 15+ team members across scheduling, inventory, and customer service; implemented process optimization initiatives improving operational efficiency 20% through systematic workflow documentation and standardized procedures.
- Maintained technical documentation for operational milestones and site conditions; tracked performance metrics to identify issues; designed training protocols for team members ensuring knowledge transfer and system adoption across the organization.
- Led continuous improvement initiatives analyzing workflow data to identify bottlenecks; implemented solutions reducing operational costs while maintaining service quality standards and team performance metrics.

President, Automotive Club, Diablo Valley College — *Oct 2025 – May 2026*
Electrical Systems Leadership — Concord, CA

- Led hands-on electrical systems analysis and power characterization across vehicle diagnostic projects for 50+ members; developed technical testing procedures and electrical system validation protocols ensuring consistent project execution.
- Engineered comprehensive technical documentation including electrical system specifications, circuit interpretation frameworks, and equipment procedures; implemented standardized protocols for system diagnostics and performance validation.
- Coordinated hands-on training sessions on vehicle electrical diagnostics, oscilloscope measurements, and embedded system troubleshooting; mentored club members in systematic debugging and fostered collaborative learning environment that increased technical competency.

PROJECTS

Power Distribution Failure Analysis (Completed)

- Built NetworkX-based grid simulator modeling substations and transmission lines; implemented DC power flow calculations using susceptance matrix and linear algebra.
- Developed cascade detection algorithm identifying cascading failures when lines overload; simulates fault injection and analyzes grid resilience under critical component failures.
- Demonstrates power systems optimization relevant to utility infrastructure planning and reliability engineering

Smart Microgrid Simulator (In Progress, Oct 2026)

- Arduino Mega-based real-time monitoring: voltage/current sensors across solar, battery, and grid sources with relay-based switching and DHT22 temperature measurement.
- Python Flask backend: power flow optimization algorithm prioritizing solar→battery→grid to minimize cost and emissions; Socket.io live data streaming and SQLite logging.
- React dashboard: real-time visualization with Charts.js showing power/voltage/current/temperature alongside animated power flow diagram; full-stack IoT for career fair demo.

TECHNICAL SKILLS & TOOLS

Power Systems & Electrical Engineering: Power systems modeling and analysis, DC/AC power flow calculations, grid topology design, susceptance matrix methods, fault analysis and cascade detection, power distribution optimization, grid resilience evaluation, microgrids and renewable integration, electrical specifications and power calculations, utility infrastructure planning

Programming Languages & Frameworks: Python (NumPy, SciPy, NetworkX, Pandas, Flask, Socket.io, PySerial, Jupyter), C++, MATLAB (statistical analysis, signal processing, data visualization), JavaScript, Node.js, Express.js, React (Charts.js, Recharts, component architecture, state management), HTML/CSS, SQLite (database design and queries), RESTful API design and development, real-time data streaming and websockets

Circuit Design & Simulation: LTSpice circuit simulation and power analysis, resonant circuit design, power supply design, component selection and power calculations, KiCAD schematic capture, PCB layout and design, board-level analysis and troubleshooting, SolidWorks CAD modeling

Hardware & Embedded Systems: Arduino Mega development and firmware programming, sensor integration (voltage sensors, current sensors, temperature sensors, relay modules), real-time data acquisition and analog-to-digital conversion, hardware-software integration, protocol implementation (Serial, Socket.io, REST APIs)

Test Equipment & Instrumentation: Oscilloscope proficiency (AC/DC signal measurement, waveform analysis, frequency response, circuit debugging), multimeter operation (voltage/current/resistance measurement), component-level testing and validation, breadboard prototyping, hands-on circuit testing and measurement